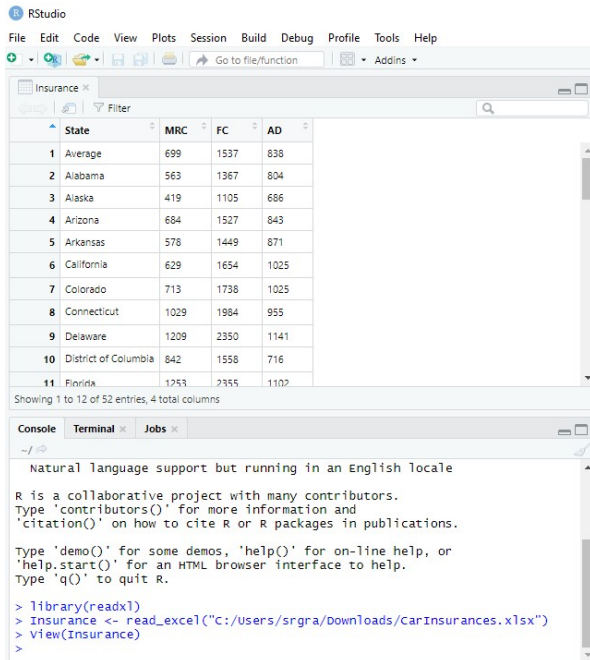


Sean Graham Week 1

1.



RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function Addins

Insurance

	State	MRC	FC	AD
1	Average	699	1537	838
2	Alabama	563	1367	804
3	Alaska	419	1105	686
4	Arizona	684	1527	843
5	Arkansas	576	1449	871
6	California	629	1654	1025
7	Colorado	713	1738	1025
8	Connecticut	1029	1984	955
9	Delaware	1209	2350	1141
10	District of Columbia	842	1558	716
11	Florida	1253	2355	1102

Showing 1 to 12 of 52 entries, 4 total columns

Console Terminal Jobs

```
~/
Natural language support but running in an English locale

R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

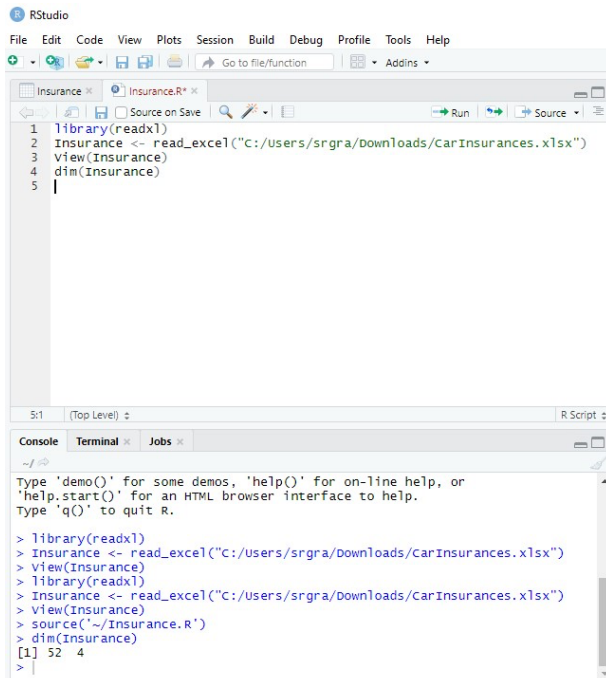
Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

> library(readxl)
> Insurance <- read_excel("C:/Users/srgra/Downloads/CarInsurances.xlsx")
> view(Insurance)
>
```

2 and 3.

There are 52 rows, representing the row naming the columns, the 50 states, and the District of Columbia

There are four columns, representing the states, the MRC (average annual price of minimum required coverage), the FC (average annual price of full coverage) and the AD (the annual difference between the MRC and FC).



RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function Addins

Insurance Insurance.R

```
1 library(readxl)
2 Insurance <- read_excel("C:/Users/srgra/Downloads/CarInsurances.xlsx")
3 view(Insurance)
4 dim(Insurance)
5 |
```

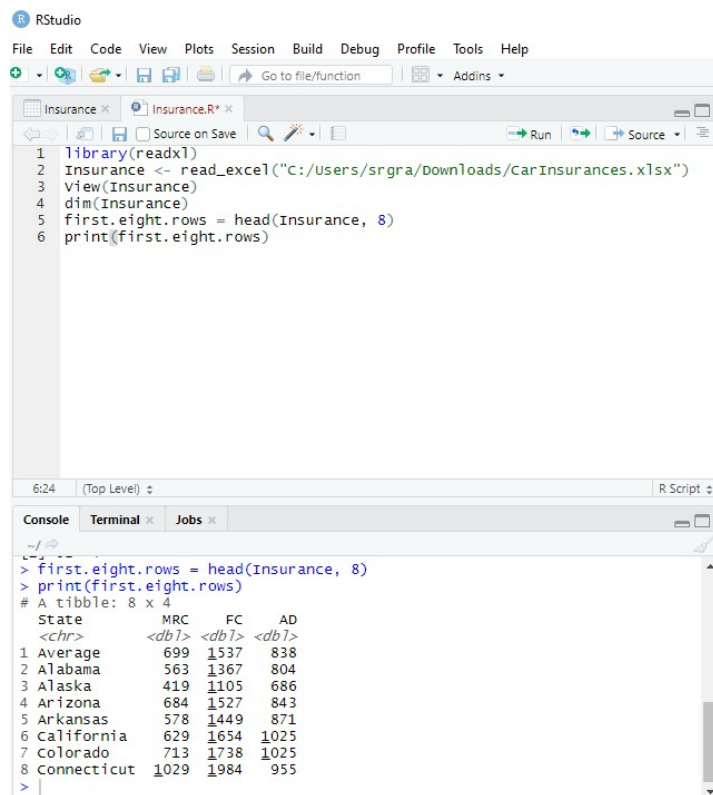
5:1 (Top Level) R Script

Console Terminal Jobs

```
~/
Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

> library(readxl)
> Insurance <- read_excel("C:/Users/srgra/Downloads/CarInsurances.xlsx")
> view(Insurance)
> library(readxl)
> Insurance <- read_excel("C:/Users/srgra/Downloads/CarInsurances.xlsx")
> view(Insurance)
> source("~/Insurance.R")
> dim(Insurance)
[1] 52 4
>
```

4.



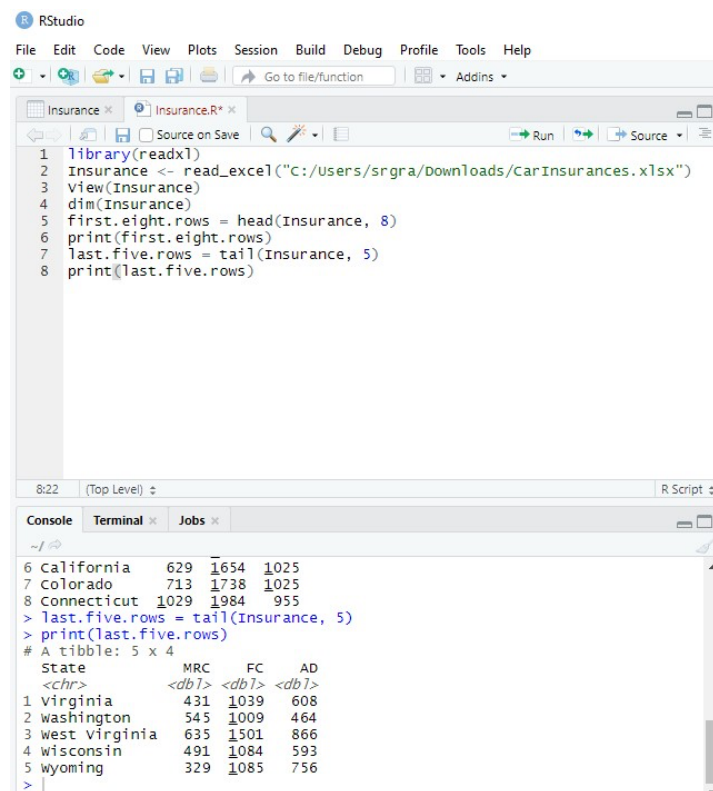
The RStudio interface shows the following code in the script editor:

```
1 library(readxl)
2 Insurance <- read_excel("C:/Users/srgra/Downloads/CarInsurances.xlsx")
3 view(Insurance)
4 dim(Insurance)
5 first.eight.rows = head(Insurance, 8)
6 print(first.eight.rows)
```

The console output shows the result of the code execution:

```
> first.eight.rows = head(Insurance, 8)
> print(first.eight.rows)
# A tibble: 8 x 4
  State      MRC    FC    AD
<chr>    <dbl> <dbl> <dbl>
1 Average    699   1537   838
2 Alabama    563   1367   804
3 Alaska     419   1105   686
4 Arizona    684   1527   843
5 Arkansas   578   1449   871
6 California 629   1654   1025
7 Colorado   713   1738   1025
8 Connecticut 1029  1984   955
```

5.



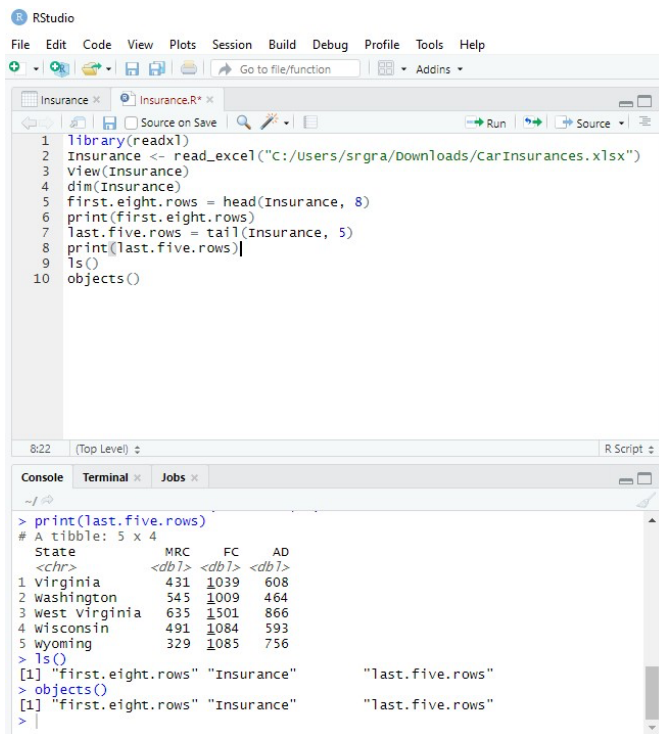
The RStudio interface shows the following code in the script editor:

```
1 library(readxl)
2 Insurance <- read_excel("C:/Users/srgra/Downloads/CarInsurances.xlsx")
3 view(Insurance)
4 dim(Insurance)
5 first.eight.rows = head(Insurance, 8)
6 print(first.eight.rows)
7 last.five.rows = tail(Insurance, 5)
8 print(last.five.rows)
```

The console output shows the result of the code execution:

```
6 California    629   1654   1025
7 Colorado     713   1738   1025
8 Connecticut  1029  1984    955
> last.five.rows = tail(Insurance, 5)
> print(last.five.rows)
# A tibble: 5 x 4
  State      MRC    FC    AD
<chr>    <dbl> <dbl> <dbl>
1 Virginia    431   1039   608
2 Washington  545   1009   464
3 West Virginia 635   1501   866
4 Wisconsin   491   1084   593
5 Wyoming     329   1085   756
```

6.



```
1 library(readxl)
2 Insurance <- read_excel("c:/Users/srggra/Downloads/CarInsurances.xlsx")
3 view(Insurance)
4 dim(Insurance)
5 first.eight.rows = head(Insurance, 8)
6 print(first.eight.rows)
7 last.five.rows = tail(Insurance, 5)
8 print(last.five.rows)
9 ls()
10 objects()
```

Console output:

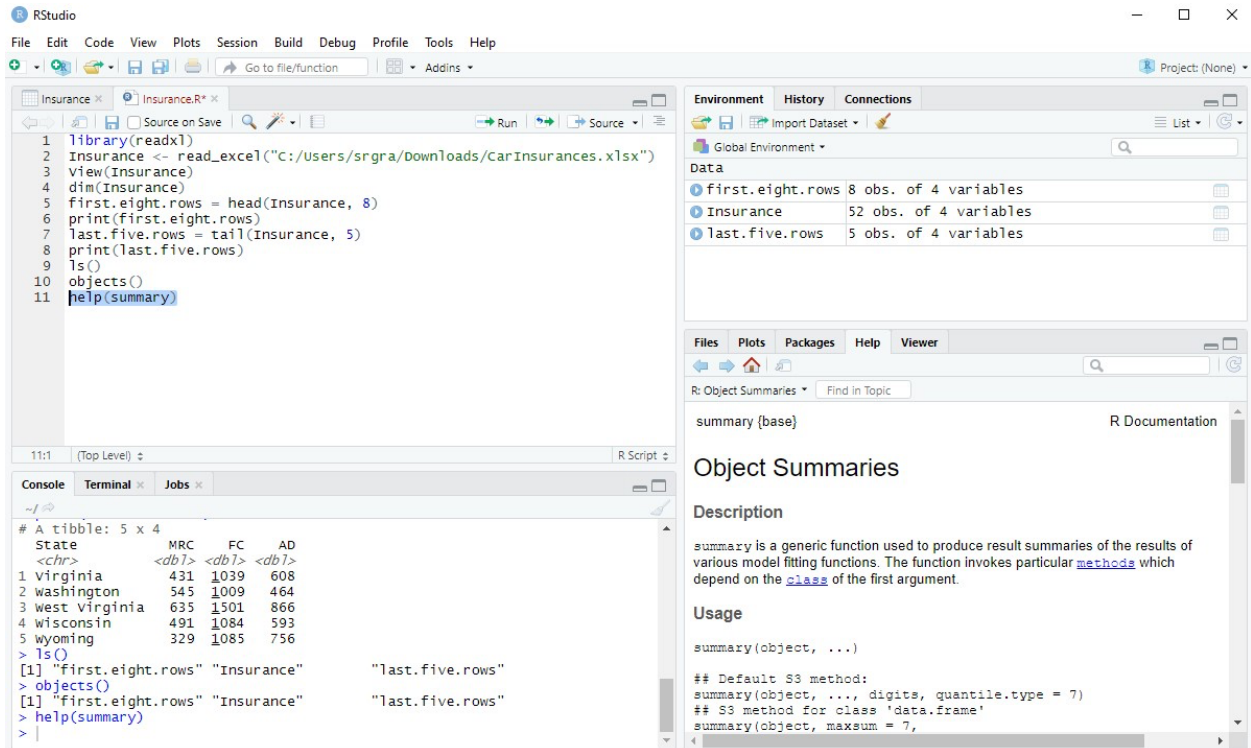
```
> print(last.five.rows)
# A tibble: 5 x 4
  State      MRC    FC    AD
  <chr>    <dbl> <dbl> <dbl>
1 Virginia  431  1039  608
2 Washington 545  1009  464
3 West Virginia 635  1501  866
4 Wisconsin  491  1084  593
5 Wyoming   329  1085  756

> ls()
[1] "first.eight.rows" "Insurance"      "last.five.rows"

> objects()
[1] "first.eight.rows" "Insurance"      "last.five.rows"

>
```

7. (help function)



```
1 library(readxl)
2 Insurance <- read_excel("c:/Users/srggra/Downloads/CarInsurances.xlsx")
3 view(Insurance)
4 dim(Insurance)
5 first.eight.rows = head(Insurance, 8)
6 print(first.eight.rows)
7 last.five.rows = tail(Insurance, 5)
8 print(last.five.rows)
9 ls()
10 objects()
11 help(summary)
```

Environment:

Object	Size
first.eight.rows	8 obs. of 4 variables
Insurance	52 obs. of 4 variables
last.five.rows	5 obs. of 4 variables

R Documentation: summary (base)

Object Summaries

Description

summary is a generic function used to produce result summaries of the results of various model fitting functions. The function invokes particular [methods](#) which depend on the [class](#) of the first argument.

Usage

```
summary(object, ...)
```

Default S3 method:

```
summary(object, ..., digits, quantile.type = 7)
## S3 method for class 'data.frame'
```

```
summary(object, maxsum = 7,
```

7. (questions)

a. The mean of MRC is 700.2

b. The mean of FC is 1539

c. The mean of AD is 838.6

The screenshot shows the RStudio interface with the following components:

- Source Editor:** Contains an R script for loading and summarizing an Excel file.
- Console:** Shows the execution of the script, including the output of `ls()`, `objects()`, `help(summary)`, and `summary(Insurance)`.
- Environment:** Lists the objects in the global environment: `first.eight.rows` (8 obs. of 4 variables), `Insurance` (52 obs. of 4 variables), and `last.five.rows` (5 obs. of 4 variables).
- Help Viewer:** Displays the documentation for the `summary` function.

```
1 library(readxl)
2 Insurance <- read_excel("C:/Users/srgra/Downloads/CarInsurances.xlsx")
3 view(Insurance)
4 dim(Insurance)
5 first.eight.rows = head(Insurance, 8)
6 print(first.eight.rows)
7 last.five.rows = tail(Insurance, 5)
8 print(last.five.rows)
9 ls()
10 objects()
11 help(summary)
12 summary(Insurance)
```

Console Output:

```
> ls()
[1] "first.eight.rows" "Insurance"      "last.five.rows"
> objects()
[1] "first.eight.rows" "Insurance"      "last.five.rows"
> help(summary)
> summary(Insurance)
      State      MRC      FC      AD
Length:52   Min.   : 309.0   Min.   : 961   Min.   : 464.0
Class :character 1st Qu.: 492.5   1st Qu.:1106   1st Qu.: 655.0
Mode :character  Median : 598.5   Median :1453   Median : 795.5
              Mean   : 700.2   Mean   :1539   Mean   : 838.6
              3rd Qu.: 824.5   3rd Qu.:1682   3rd Qu.: 986.8
              Max.   :2696.0   Max.   :3986   Max.   :1794.0
```

Help Viewer Details:

summary

For `factor`s, the frequency of the first maximum - 1 most frequent levels is shown, and the less frequent levels are summarized in " (Others) " (resulting in at most maximum frequencies).

The functions `summary.lm` and `summary.glm` are examples of particular methods which summarize the results produced by `lm` and `glm`.

Value

The form of the value returned by `summary` depends on the class of its argument. See the documentation of the particular methods for details of what is produced by that method.

The default method returns an object of class `c("summaryDefault", "table")` which has specialized `format` and `print` methods. The `factor` method returns an integer vector.

The matrix and data frame methods return a matrix of class `"matrix"` obtained by